

|    |          |                                                                                                                  |
|----|----------|------------------------------------------------------------------------------------------------------------------|
|    |          | $g_y, g_x = 1.0/0.00 = 0.11$                                                                                     |
| 12 | <b>A</b> | Using $E_k = \frac{Gm_E m}{2r}$ , $m_A$ is larger than $m_B$ and since both satellites have same kinetic energy, |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | <p>Using total energy <math>= -\frac{Gm_E m}{2r}</math>, satellites A and B have the same total energy, (Option A is false)</p> <p>Orbital radius for satellite A is larger than that of satellite B. (Option B is true)</p> <p>Using <math>T^2</math> is proportional to <math>r^3</math>, satellite A has a larger period (Option C is true)</p> <p>Since angular velocity is inversely proportional to <math>T</math>, satellite A has smaller angular velocity (Option D is true)</p> |
| 13 | B | 1 ( 2 ) ( 5 )                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |